The Effectiveness of Macroprudential Policies in Thailand: an Empirical Analysis using Micro-Level Data

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Outline

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I. Conceptual Framework and Research Questions

The journey of Macroprudential Policy (MaPP)

- MaPP is not a new concept. The first mention of MaPP can be dated back to the 1970s.
- Only after GFC in 2009 that policymakers renewed interest in MaPP.
  An important lesson from the crisis is that regulating financial instuitions on a standalone basis is not enough to maintain stability of the whole financial system.
- MaPP addresses important drawbacks of Microprudential Policy.

	Microprudential	Macroprudential
Objectives	• Limit distress of individual institutions • Consumer protection	• Limit financial system-wide distress • Avoid macroeconomic costs limked to financial stability
Characterization of Risk	Exogenous (independent of individual agents' behavior)	Endogenous (dependent on collective behavior)
Correlations and common exposure across institutions	Irrelevant	Important
Calibration of prudential controls	In terms of risks of individual institutions	In terms of system-wide risk

The journey of Macroprudential Policy (MaPP)

❖ Following the GFC have seen a greater adoption of MaPP across regions, with a variety of tools.

Risk factors	Risk dimensions		Example of MaPP measures
Risk buildup from economic activities	Credit side	Broad-based tools	**Dynamic provisioning requirements,** countercyclical capital buffer, (time-varying) leverage ratio cap, caps on credit growth
		Sectoral tools	**Loan-to-value ratios**, debt-service ratios, **debt-to-income ratios**, sectoral capital requirements, exposure limits
Spillover risks from the financial system structure	Structural side	Systemically Important Financial Institutions (SIFIs)	**Capital surcharge on Domestic/Global Systemically Important Banks (D-SIBs/G-SIBs)**
Risks from inappropriate Asset-Liability Management	Liquidity	- Liquidity mismatch - Maturity mismatch - Currency mismatch	Loan to deposit ratio, reserve requirements, levy on non-core funding, limit on net open FX position
Risks from excessive financial market volatility	Financial market stability	- Impact from excess volatility in the financial market	Capital flows management measures for MaPP purposes, regulations on investment and leverage

Research about MaPP Effectiveness

- The greater adoption has triggered research interest in effectiveness of MaPP.
- Current studies to evaluate the effectiveness of MaPP can be divided into two strands – cross-country analysis using aggregate data, and individual-country analysis using micro-level data.
- Evidence has been mixed.
  - Inconclusive results in part reflect a great variety of MaPP tools being applied in different institutional setups and time periods.
- Research has geared toward finding evidence from micro-level data to disentangle loan demand and supply, to focus on the specific sectors, and to investigate response among different players.

Our Research Questions

- **Does MaPP in Thailand achieve its objectives? Do banks respond differently to the policy?**
  - Policy effectiveness needs to be evaluated against the intended impact set by policymakers prior to policy enforcement. The micro-level data allows us to do this.
  - Our study abstracts away from the questions of
    - Interaction between MaPP and Monetary Policy
    - Possible leakgaes to the unregulated financial sector/ cross-border transaction

II. MaPP in Thailand

MaPP in Thailand

- Like other emerging-market central banks, the BOT has been aware of the need to employ MaPP in containing systemic risks, induced by lessons learnt from previous crises.
- Since 2003, the BOT has implemented four forms of MaPP
  - **LTV measures**
  - **Dynamic loan-loss provisioning**
  - **Maximum credit limits on credit cards and personal loans**
  - **Capital surcharge for five SIBs**
- In this study, we will focus on the LTV measures introduced in 2009, 2011 and 2013.

The LTV measures in Thailand

- The primary aim of LTV measures is to ensure that banks are sufficiently prudent in their lending standard. Impact on credit growth is a welcoming effect, but of secondary importance.
  - The most direct way to measure policy effectiveness is LTV-setting behavior of banks.
- The LTV measures were triggered based on detection of systemic risks. Each measure targets specific loan sectors.
- Strict maximum limit and then risk-weighted approach have been applied.

The LTV measures in Thailand (2)

Measures	Motivation	Target	Action	Status
2003	• Prevent risk build-up in the property market	Mortgage loan of **properties ≥ 10 million baht (HV)**	Impose strict LTV limit at 70%	removed
2009	Support activities in the property market amid global economic slowdown	Mortgage loan of **properties ≥ 10 million baht (HV)**	• Increase the limit from 70% to 80%. • For LTV within the limit, risk-weighted capital charge = 35% • If LTV exceeds the limit, risk-weighted capital charge = 75%	in place

The LTV measures in Thailand (3)

Measures	Motivation	Target	Action	Status
2011	Prevent build-up of risks in the property market	Mortgage loan of **high-rise properties < 10 million baht (HR-LV)**	• Impose the limit at 90% • The same capital-surcharge rule as 2009	in place
2013	• Prevent build-up of risks in the property market • Originally planned to enforce in 2012, but postponed to 2013 due to flood	Mortgage loan of **low-rise properties < 10 million baht (LR-LV)**	• Impose the limit at 95% • The same capital-surcharge rule as 2009	in place

III. Data and Methodology

Data

- Two sets of data are employed for our study: Mortgage Loan Database (MGL) and Banks' balance sheet
- **MGL**
  - Records amount of property-collateralized loans newly issued at the contract level. The data contains associated borrower- and loan characteristics.
  - Includes all commercial banks operating in Thailand, including foreign branches and subsidiaries (23 banks in total)
  - Quarterly basis from 2007Q1 to 2017Q3
- **Banks' balance sheets**
  - We use information on bank characteristics (e.g. asset sizes, liquidity ratio) and amount of loan outstanding (for credit-growth calculation) for this analysis.
  - Bank coverage matches MGL data.
  - Quarterly data from 2004Q1 to 2017Q3

Data

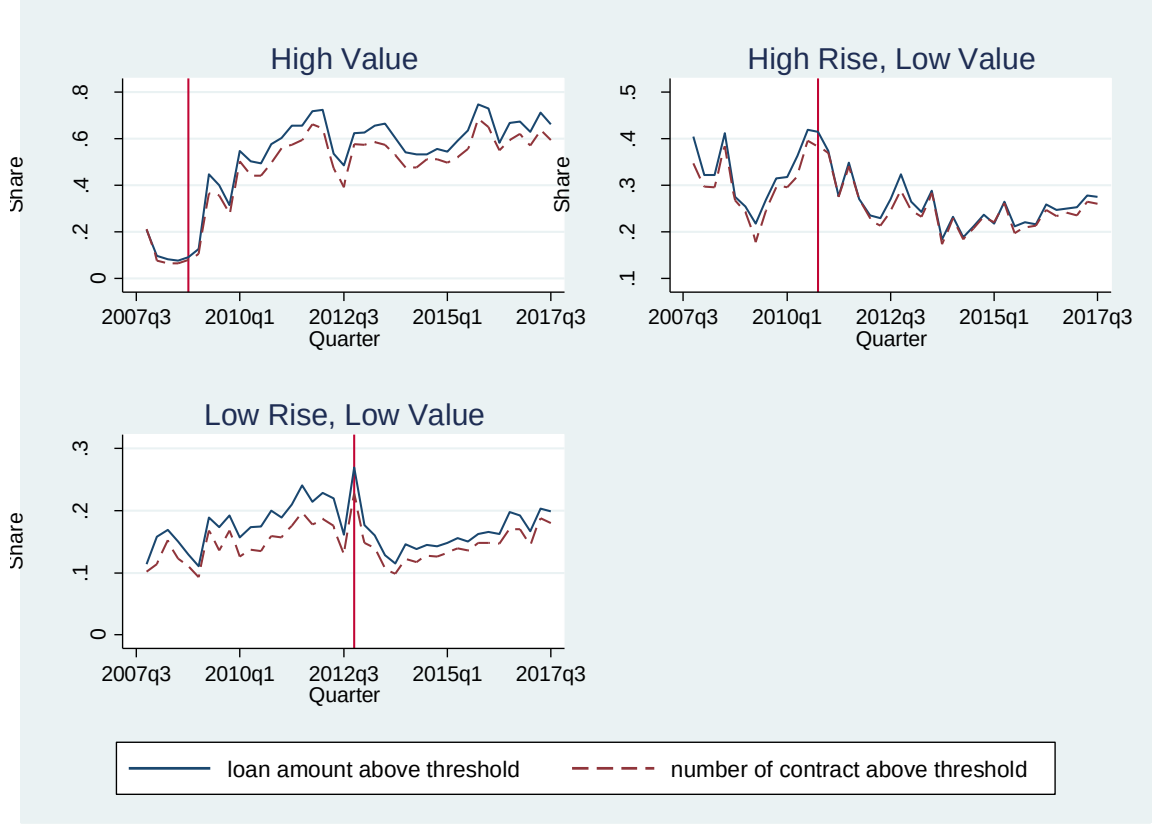
- **Advantages of using contract-level MGL data**
  - Offer new perspective in policy evaluation (LTV of each loan contract) in addition to credit growth
  - Enable clear separation of loan types, not feasible under bank-level loan data
  - Shed light on banks' different response
- **Limitations**
  - Cannot evaluate the MaPP impact on credit growth as we do with LTV-setting decision
    - The MGL records new amount of loan, not outstanding.
    - Bank-balance sheets offer loan outstanding, but types of loans cannot be divided in accordance with policy target.

Stylized Fact 1

Each LTV measure was associated with the change in the LTV distribution of mortgage loan.

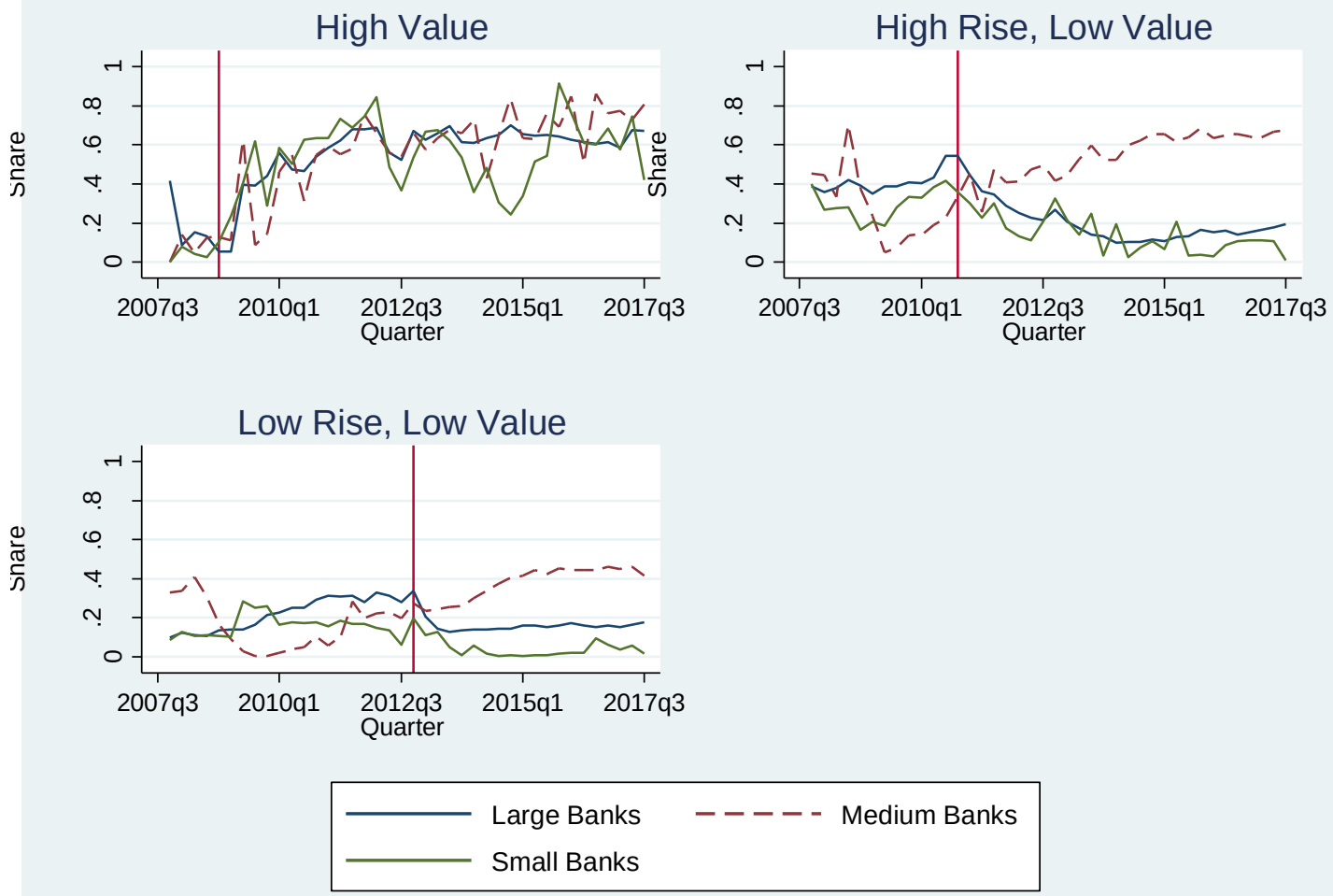
Main variable of interest:

Share of loan above threshold = total new loan above threshold/total new loan



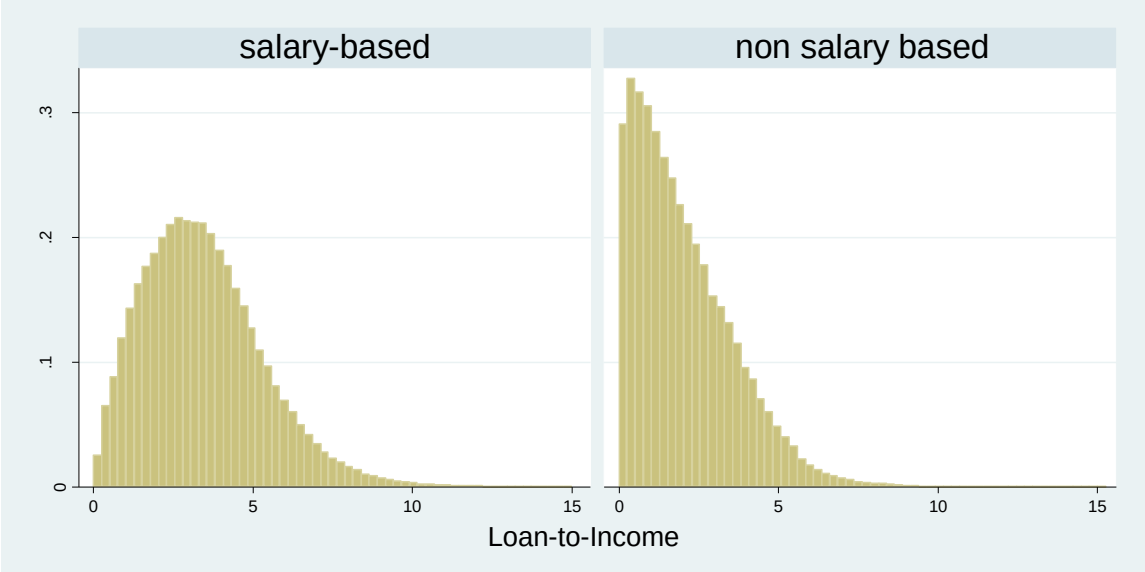
Stylized Fact 2

Different types of banks respond to LTV measures differently.



Stylized Fact 3

The LTV and LTI ratios of issued loans depend on borrowers' characteristics.



Stylized Facts 3

The LTV and LTI ratios of issued loans depend on borrowers' characteristics.

Share of LTV above threshold, by borrowers' occupation

Methodology

- We use **difference-in-difference approach (DID)** to evaluate the impact of *each LTV* measure on banks' LTV-setting behavior.
  The baseline specification consists of
  - **Dependent Variable**: share of loans above threshold
  - **Treatment group:** loan sector that the measure targets
    Control group: other types of loan
  - **Pre- and post policy intervention** is captured by a dummy variable, *LTV policy*.
  - **Other explanatory variables:** bank characteristics, macroeconomic controls, other policy controls

$$\boldsymbol{Share\ of\ loans\ above\ threshold_{b,l,t}}$$
$$= Constant + \boldsymbol{\delta_1}\ LTV_t + \boldsymbol{\delta_2}\ Target\ loan_l + \boldsymbol{\beta_1}\ LTV_t * Target\ loan_l$$
$$+ \boldsymbol{\theta'}\ Other\ LTV\ policy_t + \boldsymbol{\alpha'}\ Bank\ characteristics_{b,t} + \boldsymbol{\gamma'}\ Macro\ controls_t + \varepsilon_{b,l,t}$$

- We are interested in β1. For policy tightening (loosening), β1 should be negative (positive).
- Data is in *bank, type of loan, time* dimension (BLT).

Methodology

- Later, the baseline model is extended by
  - Including an interaction term with banks' size to allows for banks' different response and
  - Changing the dependent variable to credit growth

Methodology

- As a robustness check to the DID analysis, we perform (1) Bank-Level Regression by Loan Type and (2) Contract-Level Probit Model
- **Bank-Level Regression by Loan Type**
  - The MGL data is disaggregated into three separate datasets corresponding to the targeted sectors (HV, HR-LV, LR-LV)
  - For each of these three sets of data, we perform OLS regression to see if share of loan above LTV threshold changes after LTV measure is enforced, and if there exists variation in banks' response.
  - Data is now in *bank,time* dimension (BT).
- **Contract-level Probit Model**
  - This is to assess whether LTV measures change the probability that the loan is granted with LTV above threshold.
  - All contracts are given equal weight, *regardless of their loan amount.*
  - Borrowers' characteristics can be added into this model.

IV. Main findings

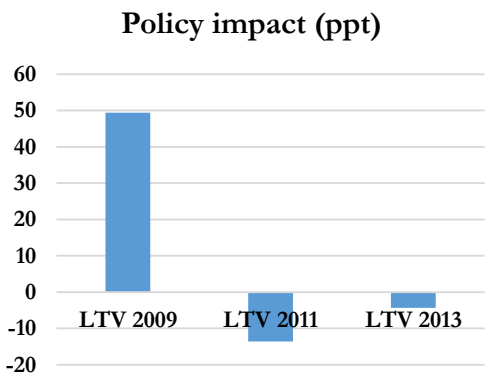
Finding 1

The LTV measures were effective in reshaping banks' LTV distribution of the targeted mortgage loans.

- **Banks responded to LTV 2009 with eagerness, indicating that the limit at 70% was binding for banks.**
- **The impact of LTV 2011 is greater than LTV 2013 due to (1) the announcement effects (2) the level of the thresholds and (3) the nature of the targeted housing sector**

Dep Var: Share of loan subject to LTV threshold	LTV above 70	LTV above 90	LTV above 95
LTV Policy	**2009**	**2011**	**2013**
HV loan	-77.313***		
	(-15.610)		
LTV 2009 x HV loan	**49.420*** **		
	(18.072)		
HR-LV loan		16.802**	
		(2.637)	
LTV 2011 x HR-LV loan		**-13.582*** **	
		(-6.934)	
LR-LV loan			6.878
			(1.832)
LTV 2013 x LR-LV loan			**-4.346** **
			(-2.710)
Observations	1523	1481	1481
R Squared	0.458	0.022	0.025

Notes (1) t-stat in parentheses (2) * p<0.05 ,** p<0.01 ,*** p<0.001 (3) some variables are omitted



Finding 2

Banks responded differently to the LTV measures

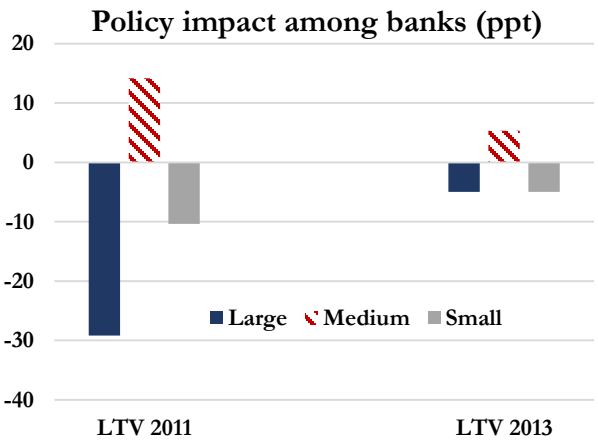
- **All measures achieve their objective on small and large banks.**
- **Medium banks behave distinctively from large and small banks, implying their competitive stance and greater willingness to bear costs.**
- **Banks that carry more housing loans in their portfolios responded more strongly to the measures.**

	(1)	(2)	(3)
	LTV 2009	LTV 2011	LTV 2013
Dep Var: Share of loan subject to LTV threshold	LTV above 70	LTV above 90	LTV above 95
LTV Policy			
LTV x Target Loan	47.849***	-10.359***	-4.992*
	(10.052)	(-3.468)	(-2.045)
LTV x Target Loan x Large bank	-5.379	-18.817***	3.233
	(-0.826)	(-4.549)	(0.975)
LTV x Target Loan x Medium bank	-0.696	24.555***	10.334**
	(-0.093)	(5.189)	(2.692)
LTV x Target Loan x High share_hloan	13.443*	-8.384*	-11.366***
	(2.099)	(-2.085)	(-3.317)
Observations	1487	1487	1487
R Squared	0.469	0.128	0.194

t statistics in parentheses

* p<0.05 ,** p<0.01 ,*** p<0.001

Note: Some variables are omitted to preserve space.



Finding 3

From borrowers' perspective, the probability of getting a loan with LTV above threshold changed after the measures were introduced.

❖ **The probit model confirms banks' differing response, found in the DID analysis.**

	(1)	(2)	(3)
Target loan subsample:	**LTV 2009**	**LTV 2011**	**LTV 2013**
Dep Var: Dummy subject to threshold	**LTV above 70**	**LTV above 90**	**LTV above 95**
Salary-based income dummy	-0.068***	0.070***	0.099***
	(-9.302)	(37.600)	(103.128)
Number of (co-)borrowers	-0.006	0.018***	0.006***
	(-1.006)	(9.351)	(7.864)
Bangkok area dummy	0.046***	0.068***	0.079***
	(5.916)	(29.558)	(90.034)
Crisis dummy	-0.198***	-0.051***	-0.031***
	(-10.829)	(-12.892)	(-13.978)
Flood dummy	0.078***	-0.049***	0.021***
	(4.873)	(-11.978)	(10.346)
LTV 2009	**0.443*** **	0.000	0.026***
	(9.472)	(0.000)	(13.683)
LTV 2011	0.036***	**-0.177*** **	0.032***
	(2.929)	**(-31.556)**	(20.970)
LTV 2013	0.053***	-0.054***	**-0.147*** **
	(5.851)	(-22.621)	**(-42.633)**
Observations	21,230	272,777	654,244

z-statistics in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Note: (1) The table reports average marginal effects
(2) Some variables are omitted to preserve space.

Finding 4

Impact of LTV policy was muted in terms of loan growth at the bank level

- **We did not find statistically significant impact of the LTV policy on the credit growth.**
- **Plausible reasons**
  - **Banks might not adjust the volume of targeted loans, but rebalanced LTV distribution of the targeted loans instead.**
  - **Alternatively, banks might change the composition of loans across the housing-loan subsectors, but maintain the same total volume of housing loans.**

	(1)	(2)	(3)
	LTV 2009	LTV 2011	LTV 2013
Dep Var: %YOY loan growth	LTV above 70	LTV above 90	LTV above 95
LTV Policy			
LTV	-0.089	-0.008	0.047
	(-0.803)	(-0.104)	(0.658)
LTV X HOUSE LOAN	**0.233***	**0.039**	**-0.155**
	(2.455)	**(0.394)**	**(-1.555)**
LTV X HOUSE LOAN X Large bank	**-0.240**	**-0.135**	**0.145**
	(-1.136)	**(-0.634)**	**(0.679)**
LTV X HOUSE LOAN X Medium bank	**-0.131**	**0.085**	**0.134**
	(-0.501)	**(0.325)**	**(0.511)**
Observations	2552	2552	2552
Overall R-Squared	0.028	0.025	0.026

t statistics in parentheses

* p<0.05 ,** p<0.01 ,*** p<0.001

Note: Some variables are omitted to preserve space.

V. Key Takeaway

- Overall, the LTV measures impact banks' lending behavior on the specific sectors they aim to target. Nevertheless, the impact does not necessarily manifest in credit growth.
- Policy design should take into account potential different response among banks.
- Central banks' warning about financial imbalances plays a part in nudging banks to adjust their lending behavior even before the policy is enforced legally. This ensures smooth transition of the policy.

Thank you